Tarea No. 3
Hugging Face Transformers

Para esta tarea cada equipo preparará y presentará en clase una de las siguientes partes de la librería de Hugging Face. Esta es una presentación un poco más detallada que les debe tomar alrededor de 30 minutos.

La repartición de los temas será como sigue:

Equipo	Partes/Temas
1	5,6
2	3,4
3	1,2

Part 1: Introduction to Hugging Face Ecosystem

Topics Covered:
- Overview of Hugging Face libraries: transformers, datasets, tokenizers
- Installation and setup (CPU/GPU)
- Pipelines for quick task execution
- Anatomy of a Hugging Face model: config, tokenizer, model

Part 2: Tokenization and Preprocessing

Topics Covered:
- Tokenization types: WordPiece, BPE, SentencePiece
- Using AutoTokenizer
- Padding, truncation, attention masks
- Preprocessing datasets using map()

Part 3: Working with Pretrained Models

Topics Covered:
- AutoModel and AutoModelFor* classes
- Forward pass and model outputs
- Model configurations and special tokens

Part 4: Fine-tuning Models on Custom Datasets

Topics Covered:
- Using Hugging Face datasets
- Trainer and TrainingArguments
- Logging and evaluation
- Saving and loading models

Part 5: Text Generation Techniques

Topics Covered:
- Greedy, beam search, top-k, nucleus (top-p) decoding
- Temperature and repetition penalty
- Prompt design

Part 6: Model Sharing, Inference, and Deployment

Topics Covered:
- Uploading to Hugging Face Hub
- Using from_pretrained with custom models
- Exporting to ONNX or TorchScript
- Serving models with FastAPI or Inference Endpoints